NATIONAL BIOINFORMATICS INFRASTRUCTURE SWEDEN

SciLifeLab

Variant-calling Workflow

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Overview

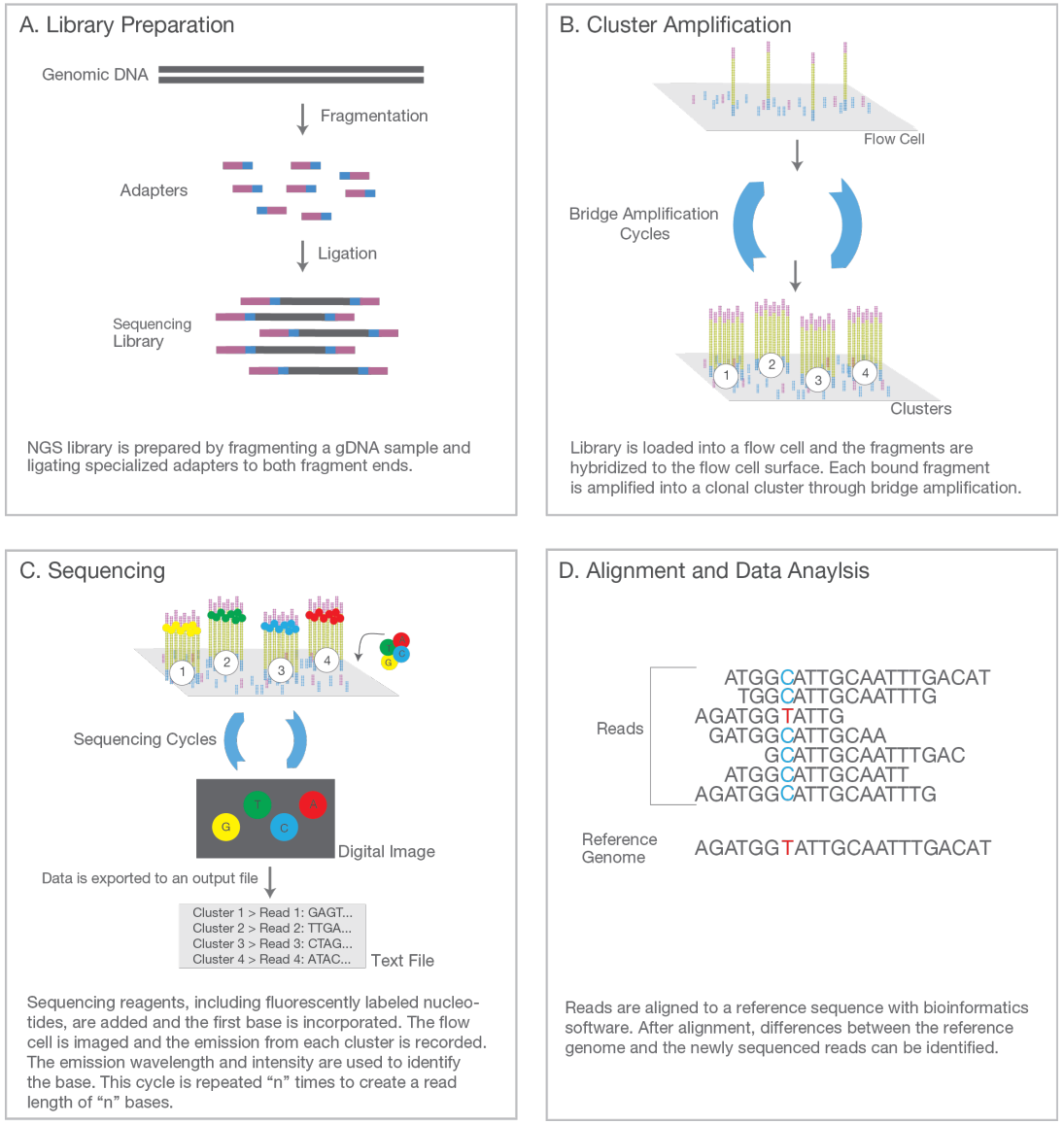
- Workflows
- Basic variant calling in one sample
- Basic variant calling in cohort
- Introduction to exercise

In separate talk Thursday at 9:

- GATK's Best practices

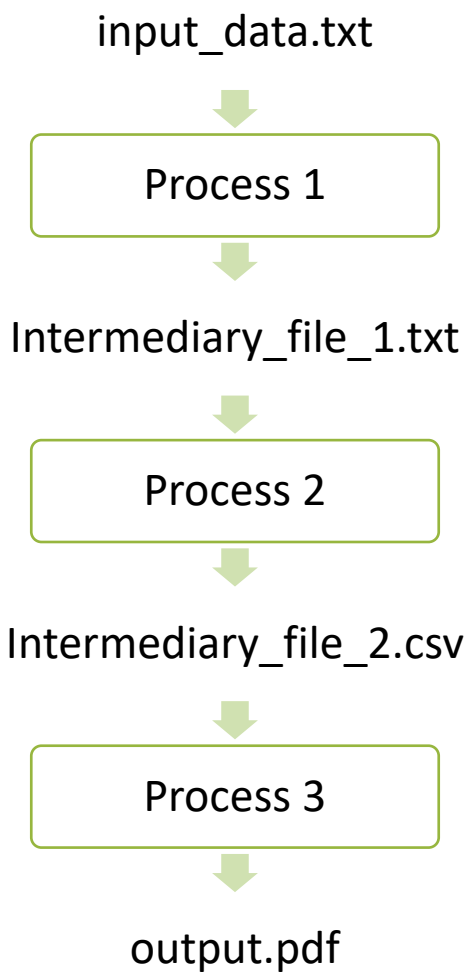
Illumina Sequencing

https://www.youtube.com/watch?v=fCd6B5HRaZ8



Workflows

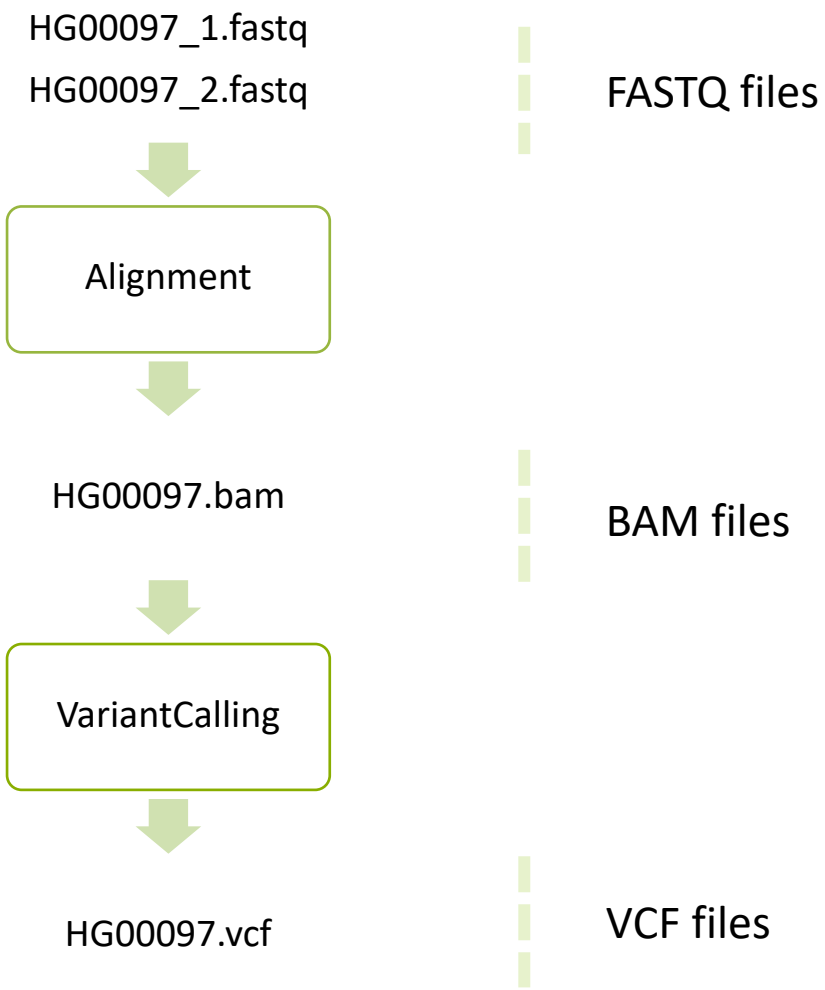
What is a workflow



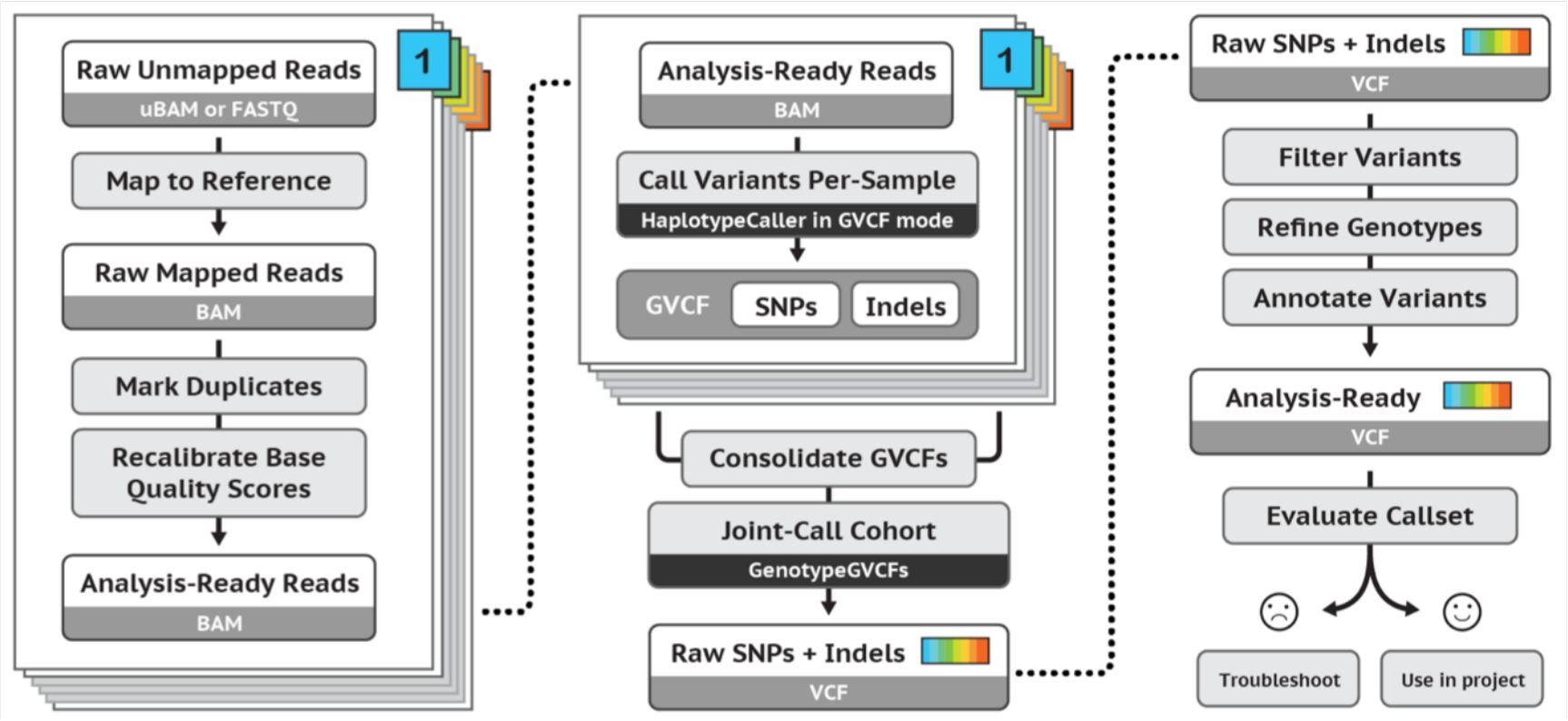
Workflow conventions

- Create a new output file in each process – don't ower write the input file
- Use informative file names
- Include information of the process in output file name

Example: Basic variant calling in one sample



GATK's best practices workflow for germline short variant discovery



https://software.broadinstitute.org/gatk/best-practices/

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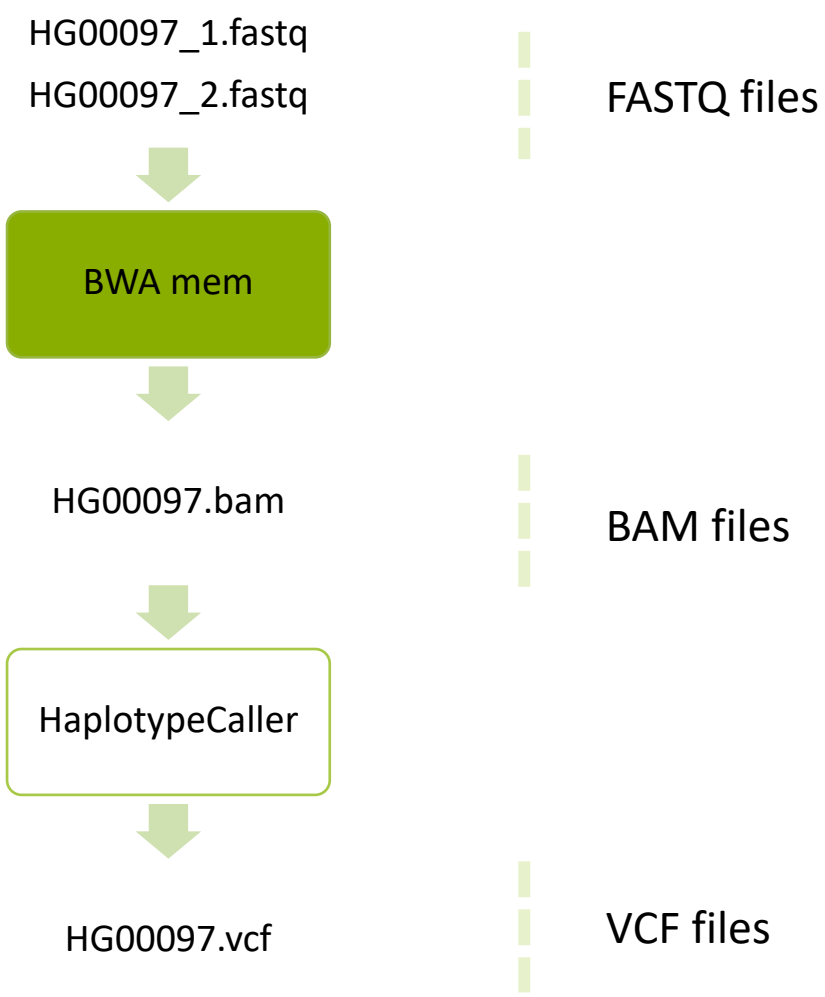
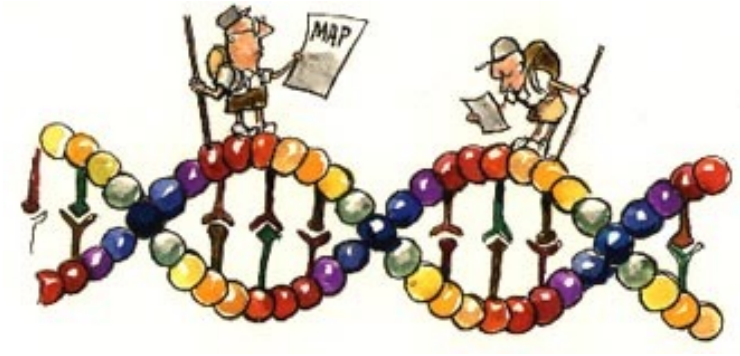
Basic Variant Calling in one sample

Alignment


HG00097_1.fastq
HG00097_2.fastq
FASTQ files
BWA mem
HG00097.bam
BAM files
HaplotypeCaller
HG00097.vcf
VCF files

The reference genome



A reference genome is a haploid nucleic acid sequence which represents a species genome.

The first draft of the human genome contained 150,000 gaps.

GRCh37: 250 gaps

GRCh38 is the latest version of the human reference genome, but we will work with GRCh37 in the lab.

Keep track of the reference version!

The reference genome sequence is used as input in many bioinformatics applications for NGS data:

- mapping
- variant calling
- annotation

You must keep track of which version of the reference genome your data was mapped to.

The same version must be used in all downstream analyses.

File indices

- Most large files we work with, such as the reference genome, need an index
- Allows efficient access to the file
- Different indices for different file-types
- Bwa index = Burrows-Wheeler transform of reference genome (several files)
- Needs index: fasta, bam vcf files

Alignment

```
module load bwa
```

```
AACAGGTATATCTTCCCCGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTACCCTCTTCCTTAGGGACTGTAC
                                  GCTAGCTAGCTAGCTACCCT
```

Burrows-Wheeler Aligner

http://bio-bwa.sourceforge.net

Burrows-Wheeler Aligner

Home

Introduction

BWA is a software package for mapping low-divergent sequences against a large reference genome, such as the human genome. It consists of three a
BWA-backtrack, BWA-SW and BWA-MEM. The first algorithm is desig
Illumina sequence reads up to 100bp, while the rest two for longer s
ranged from 70bp to 1Mbp. BWA-MEM and BWA-SW share similar fe
long-read support and split alignment, but BWA-MEM, which is the
generally recommended for high-quality queries as it is faster and m
BWA-MEM also has better performance than BWA-backtrack for 70-
reads.

FAQ

How can I cite BWA?

The short read alignment component (bwa-short) has been pub

Li H. and Durbin R. (2009) Fast and accurate short read align
Burrows-Wheeler Transform. Bioinformatics, 25:1754-60. [PM
19451168]

If you use BWA-SW, please cite:

Li H. and Durbin R. (2010) Fast and accurate long-read align
Burrows-Wheeler Transform. Bioinformatics, Epub. [PMID: 20

(See also Errata below for a minor correction to the formulae in t
papers.)

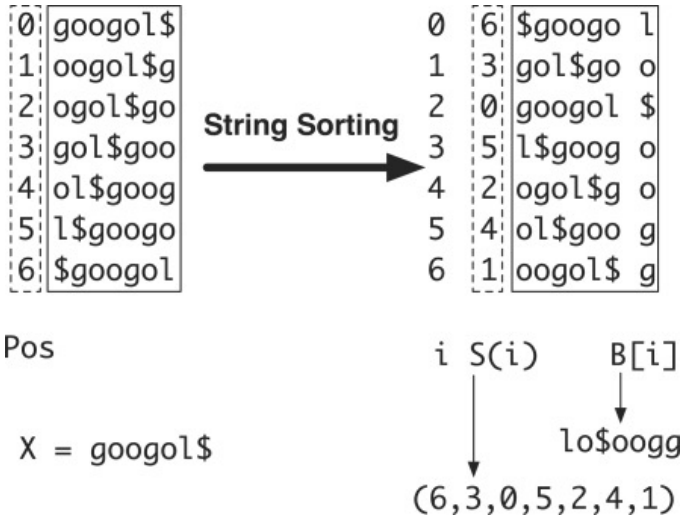
There are three algorithms, which one should I choose?

For 70bp or longer Illumina, 454, Ion Torrent and Sanger reads,
contigs and BAC sequences, BWA-MEM is usually the preferred a
For short sequences, BWA-backtrack may be better. BWA-SW ma

BWA:

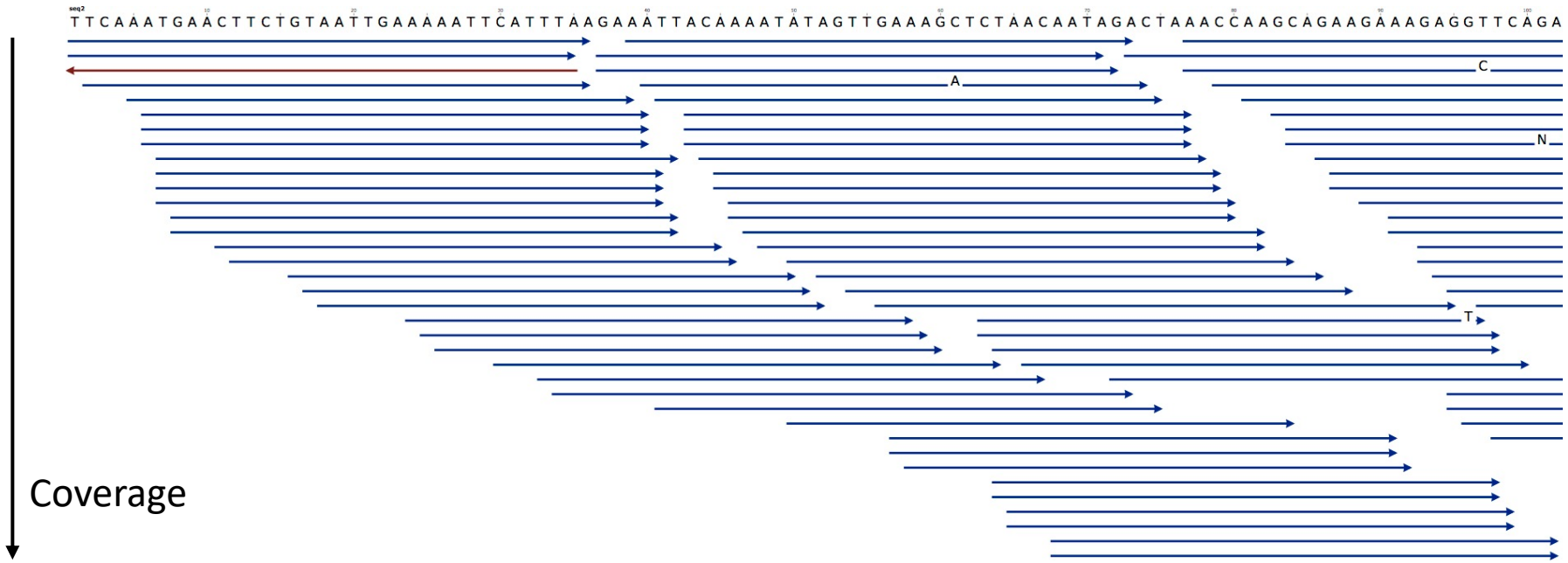
SF project page

Burrows-Wheeler transform of reference genome



Alignment

```
module load bwa
```



Output from mapping - Sam format

HEADER SECTION

```
@HD   VN:1.6SO:coordinate
@SQ   SN:2  LN:243199373
@PG   ID:bwaPN:bwaVN:0.7.17-r1188   CL:bwa mem -t 1 human_g1k_v37_chr2.fasta HG00097_1.fq HG00097_2.fq
@PG   ID:samtools PN:samtools PP:bwaVN:1.10     CL:samtools sort
@PG   ID:samtools.1     PN:samtools PP:samtools VN:1.10     CL:samtools view -H HG00097.bam
```

ALIGNMENT SECTION

```
Read_001    99    2    3843448    0    101M    =    3843625    278    TTTGGTTCCATATGAACTTT    0F<BFB<FFFBFBFFFBFBB
Read_001    147   2    3843625    0    101M    =    3843448    -278   TTATTTCATTGAGCAGTGGT    FBBI7IIFIB<BBBB<BBFF
Read_002    163   2    4210055    0    101M    =    4210377    423    TGGTACCAAAACAGAGATAT    0IIFBFFFIIIFFIFFFBBF
Read_003    99    2    4210066    0    101M    =    4210317    352    CAGAGATATAGATCAATGGA    0IIFFFIFFFIFIFIIIIIF
```

Start position

Reference sequence name

Sequence

Quality

Read name (usually more complicated)

Convert to Bam

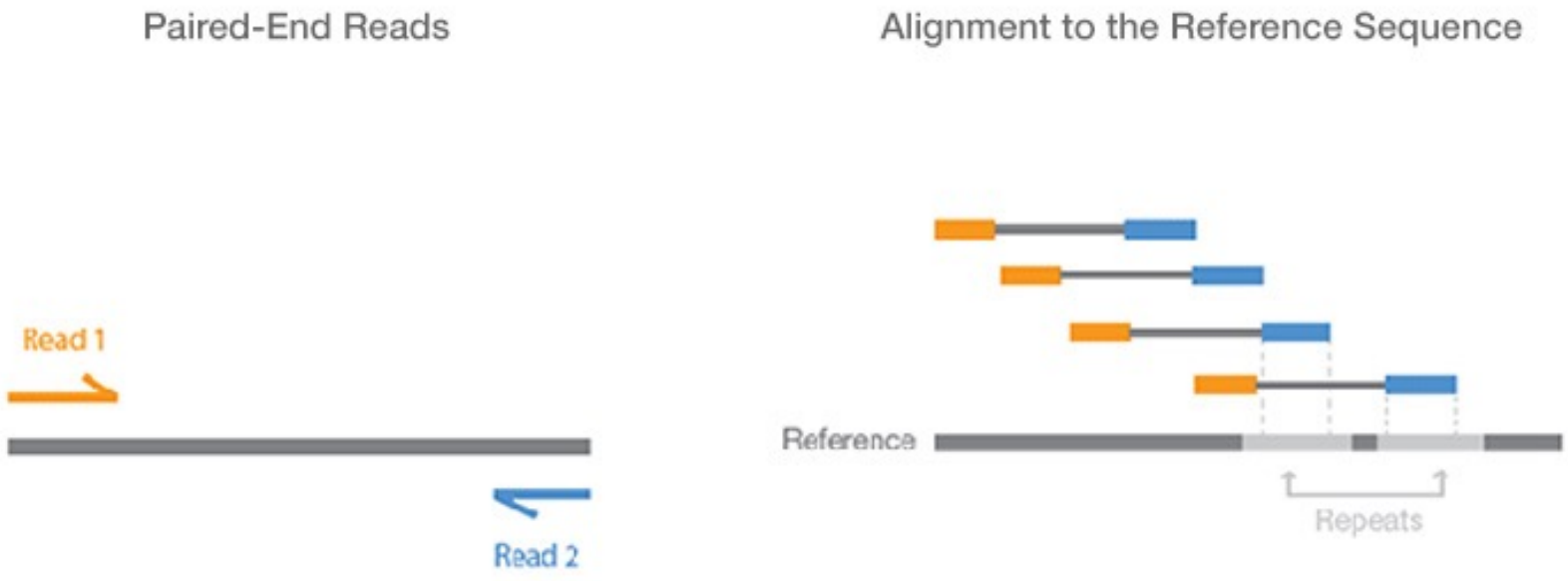
Bam file is a binary representation of the Sam file

Read groups

- Link *sample id*, *library prep*, *flowcell* and *sequencing run* to the reads.
- Good for error tracking!
- Often needed for variant calling
- Detailed description at https://gatkforums.broadinstitute.org/gatk/discussion/6472/read-groups

RGID = *combination of the sample id and run id*
RGLB = Library prep
RGPL = Platform (for us ILLUMINA)
RGPU = Run identifier *usually barcode of flowcell*
RGSM = Sample name

Paired-End data



Paired-end sequencing enables both ends of the DNA fragment to be sequenced. Because the distance between each paired read is known, alignment algorithms can use this information to map the reads over repetitive regions more precisely. This results in much better alignment of the reads, especially across difficult-to-sequence, repetitive regions of the genome.

Paired-end data

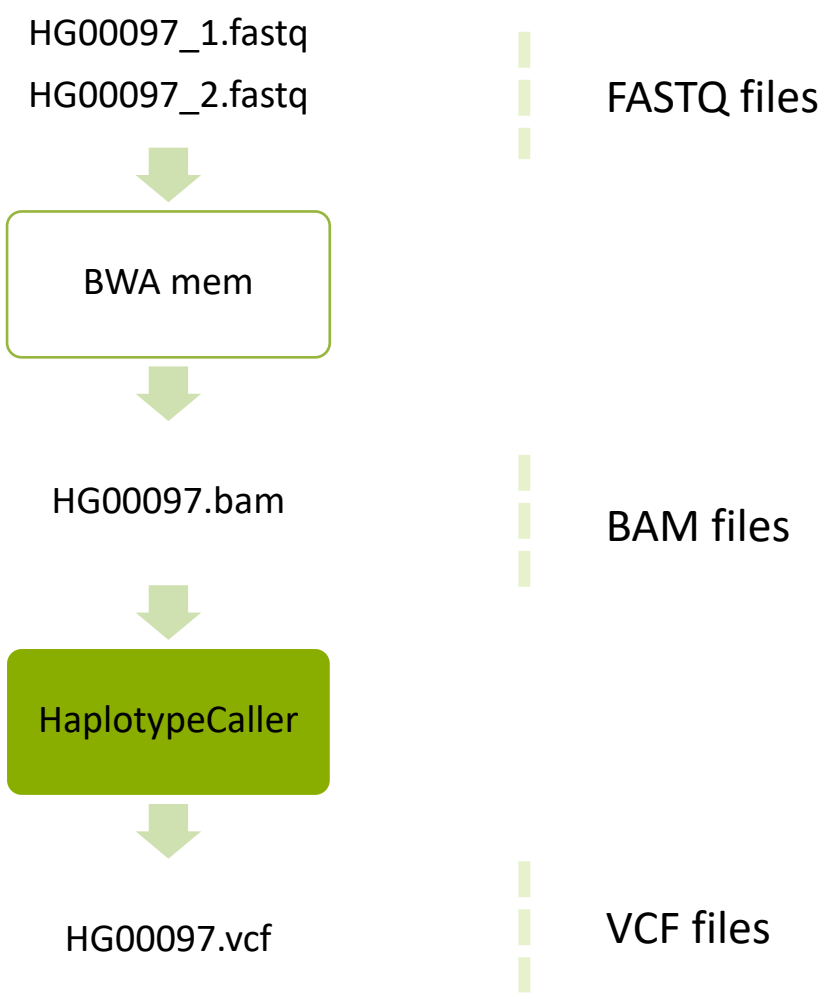
ID_R1_001.fastq

```
@HISEQ:100:C3MG8ACXX:5:1101:1160:2
197 1:N:0:ATCACG
CAGTTGCGATGAGAGCGTTGAGAAGTATAATAGG
AGTTAAACTGAGTAACAGGATAAGAAATAGTGAG
ATATGGAAACGTTGTGGTCTGAAAGAAGATGT
+
B@CFFFFFHHHHHGJJJJJJJJJJJFHHIIIIJJ
JIHGIIJJJJIJIJIJJJJIIJJJJJIIEIHHIJ
HGHHHHHDFFFEDDDDDCDDDCDDDDDDDCDC
```

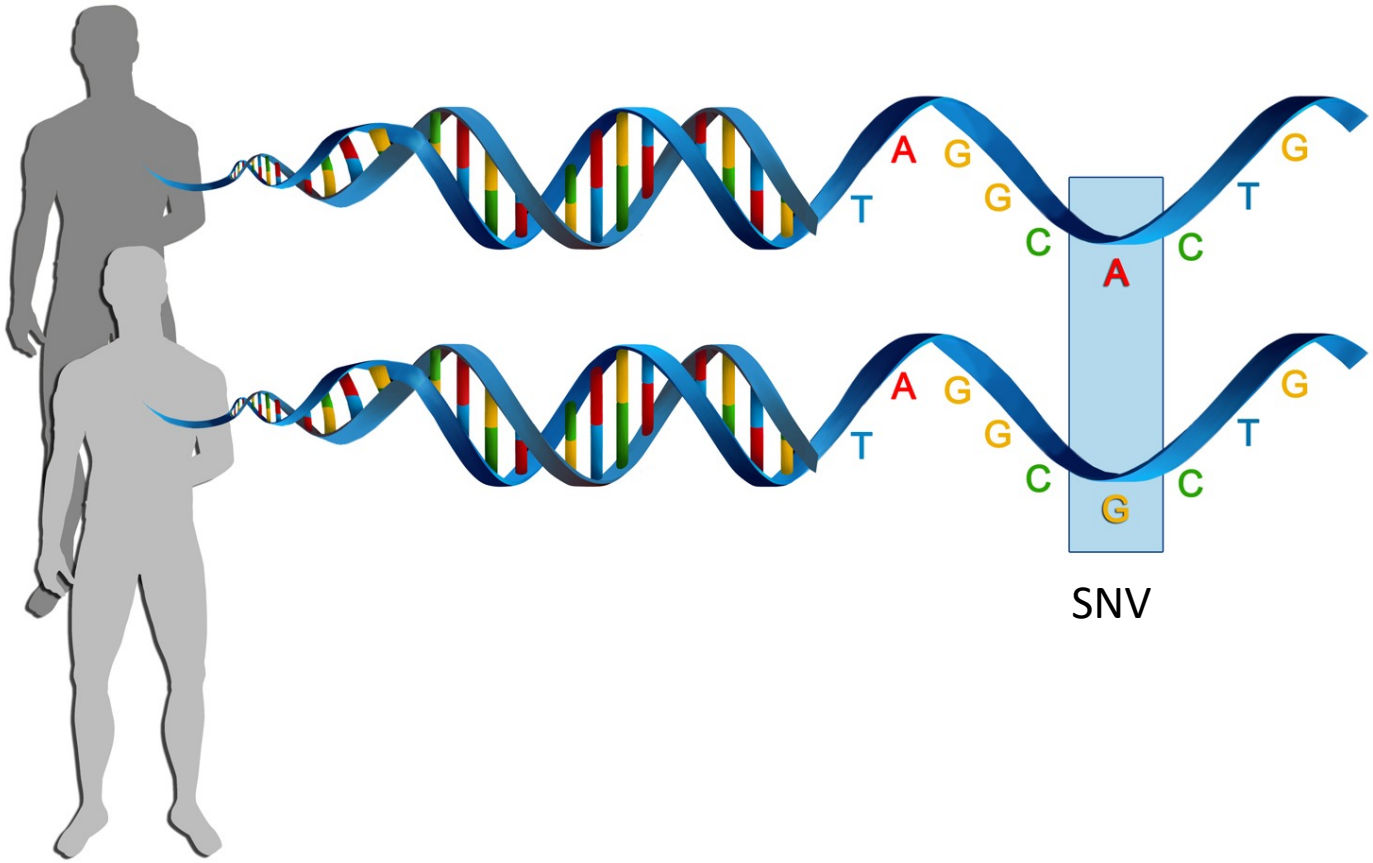
ID_R2_001.fastq

```
@HISEQ:100:C3MG8ACXX:5:1101:1160:
2197 2:N:0:ATCACG
CTTCGTCCACTTTCATTATTCCTTTCATACATG
CTCTCCGGTTTAGGGTACTCTTGACCTGGCCTT
TTTTCAAGACGTCCCTGACTTGATCTTGAAACG
+
CCCFFFFFHHHHHJJJJIJJJJJJJJJJJJJJJ
JJJJJJJIJIJGIJHBGHHIIIJIJJJJJJJJI
JJJHFFFFFFDDDDDDDDDDDDDDDDEDCCDDDD
```

Variant calling



Genetic variation



Genetic variation = differences in DNA among individuals of the same species

Detecting variants in reads

```
Reference:      ...GTGCGTAGACTGCTAGATCGAAGA...
Sample:         ...GTGCGTAGACTGATAGATCGAAGA...
                ...GTGCGTAGACTGATAGATCGAAGA...
                ...GTGCGTAGACTGCTAGATCGAAGA...
                ...GTGCGTAGACTGCTAGATCGAAGA...
                ...GTGCGTAGACTGATAGATCGAAGA...
                ...GTGCGTAGACTGATAGATCGAAGA...
                ...GTGCGTAGACTGCTAGATCGAAGA...
                ...GTGCGTAGACTGATAGATCGAAGA...
                ...GTGCGTAGACTGCTAGATCGAAGA...
                ...GTGCGTAGACTGATAGATCGAAGA...
```

Reference- and alternatve alleles

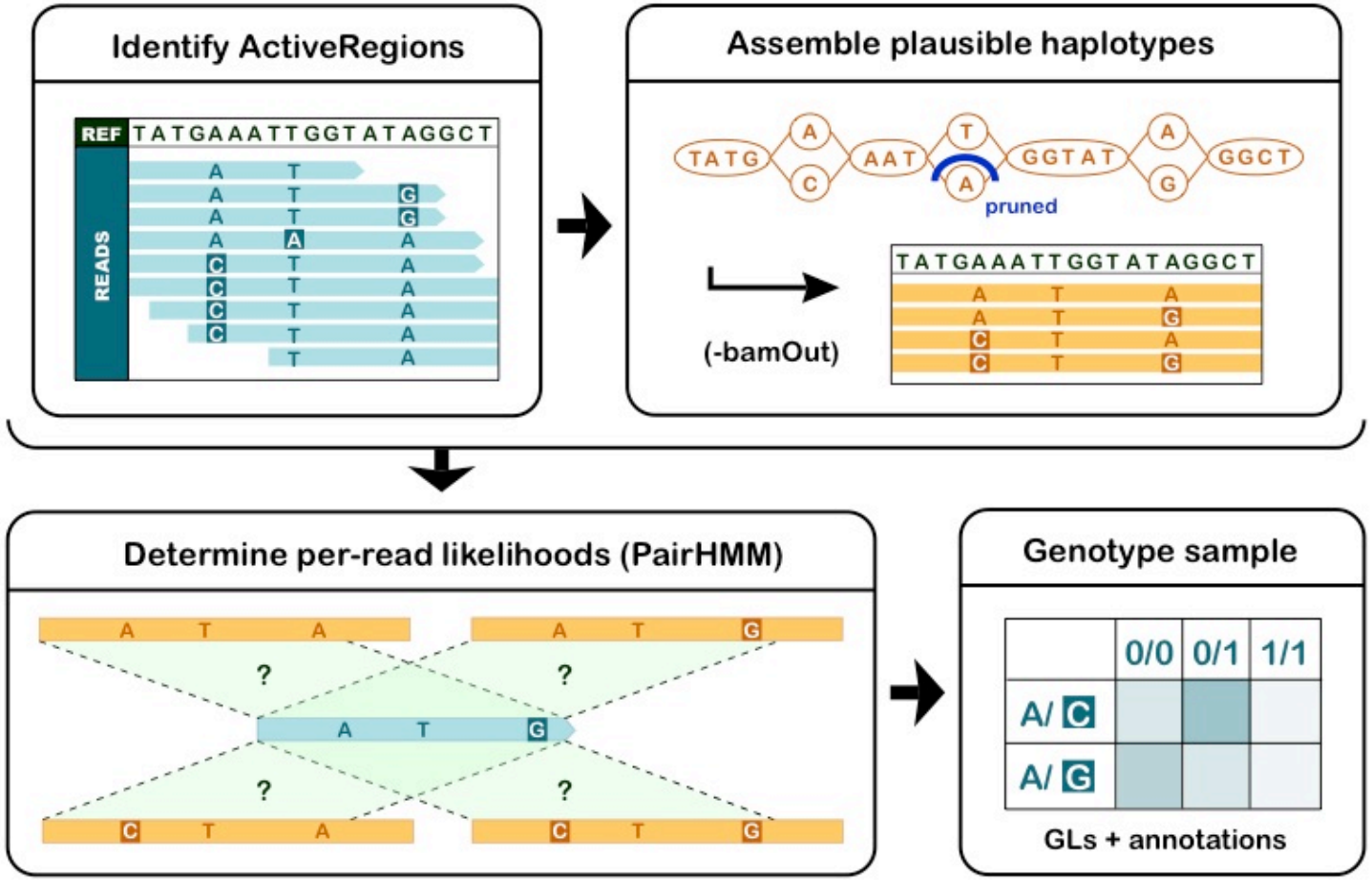
TGGGCTTTTCCAACAGGTATATCTTCCCCGCTAGCTAGCTAGCTACTTCAAATTCCT

Reference allele AGCTAGCTA

Alternative allele AGCTGGCTA

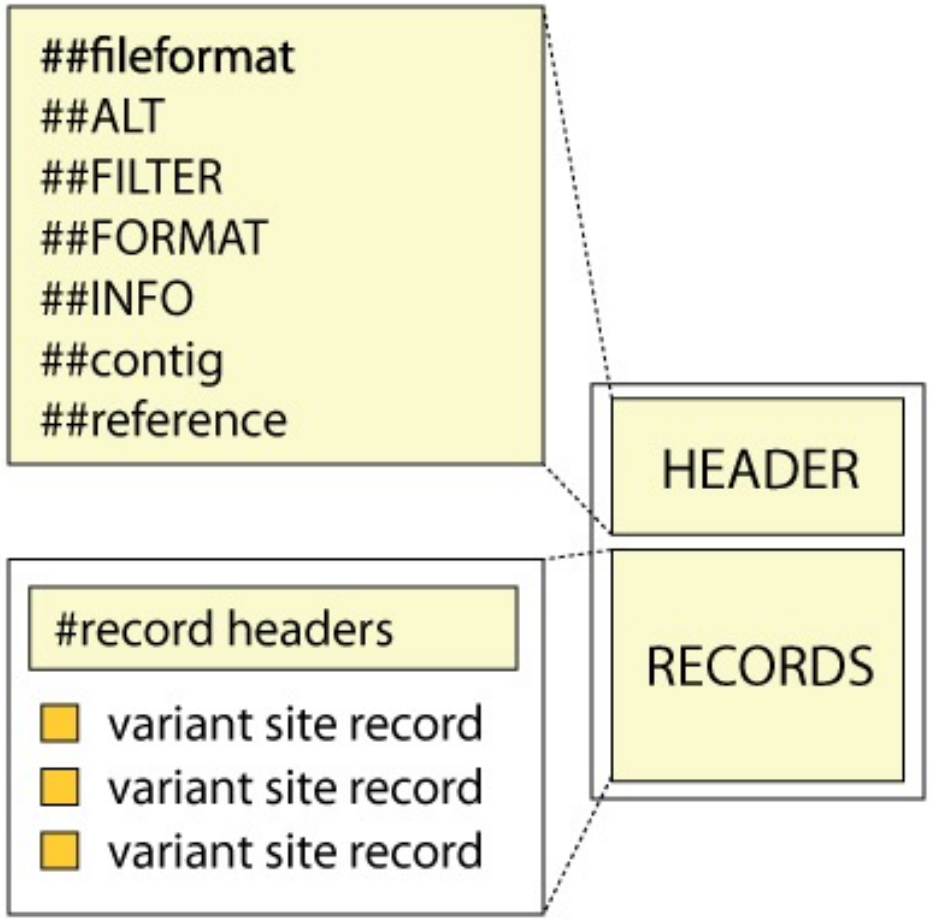
Reference allele = the allele in the refence genome
Alternative allele = the allele NOT in the refence genome

Variant Calling
HaplotypeCaller



For more info: https://www.youtube.com/watch?v=NQHGkVGICpY

Variant Call Format (VCF)



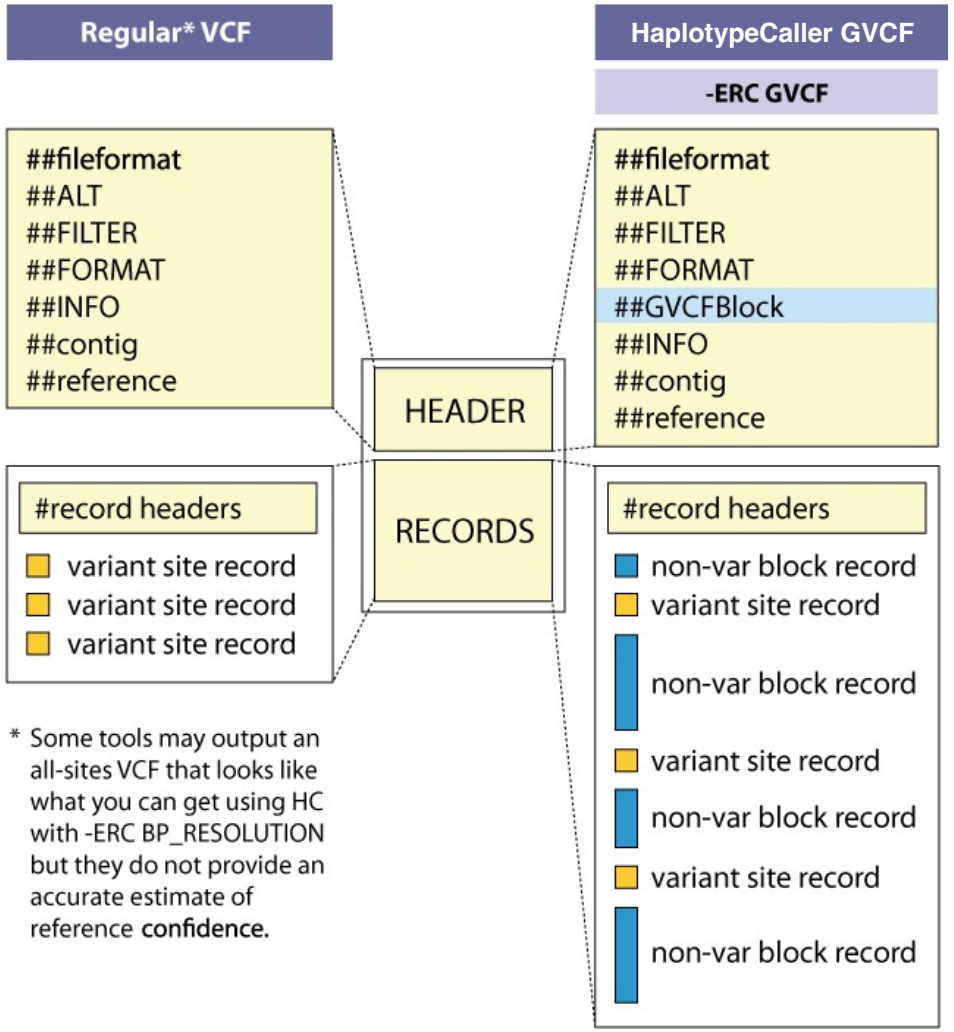
Variant Call Format (VCF)

```
##fileformat=VCFv4.2
##FILTER=<ID=LowQual,Description="Low quality">
##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Allelic depths for the ref and alt alleles in the order listed">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads with MQ=255 or with bad mates are filtered)">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes, for each ALT allele, in the same order as listed">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency, for each ALT allele, in the same order as listed">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##contig=<ID=2,length=243199373>
##source=HaplotypeCaller
#CHROM  POS          ID   REF   ALT   QUAL    FILTER   INFO              FORMAT   HG00097
2       136220992    .    G     GT    30.64   .        AC=1;AF=0.500;AN=2  GT:AD:DP  0/1:3,2:5
2       136226814    .    GAC   G     44.60   .        AC=1;AF=0.500;AN=2  GT:AD:DP  0/1:4,2:6
2       136234279    .    C     T     102.60  .        AC=1;AF=0.500;AN=2  GT:AD:DP  0/1:3,4:7
2       136234284    .    C     T     102.60  .        AC=1;AF=0.500;AN=2  GT:AD:DP  0/1:3,4:7
2       136263277    .    T     A     148.60  .        AC=1;AF=0.500;AN=2  GT:AD:DP  0/1:8,5:13
…
…
```

Variant calling in cohort

Basic variant calling in cohort

GVCF Files are valid VCFs with extra information



- GVCF has records for all sites, whether there is a variant call there or not.
- The records include an accurate estimation of how confident we are in the determination that the sites are homozygous-reference or not.
- Adjacent non-variant sites merged into blocks

Basic variant calling in cohort

Variant Call Format (VCF)

```
##fileformat=VCFv4.2
##ALT=<ID=NON_REF,Description="Represents any possible alternative allele at this location">
##FILTER=<ID=LowQual,Description="Low quality">
##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Allelic depths for the ref and alt alleles in the order listed">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads with MQ=255 or with bad mates are filtered)">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes, for each ALT allele, in the same order as listed">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency, for each ALT allele, in the same order as listed">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##contig=<ID=2,length=243199373>
##source=CombineGVCFs
##source=GenotypeGVCFs
##source=HaplotypeCaller
#CHROM  POS          ID   REF  ALT  QUAL    FILTER  INFO                FORMAT   HG00097     HG00100     HG00101
2       136045826    .    G    A    167.26  .       AC=1;AF=0.167;AN=6  GT:AD:DP 0/0:8,0:8   0/0:13,0:13 0/1:1,5:6
2       136046443    .    CGT  C    129.27  .       AC=3;AF=0.500;AN=6  GT:AD:DP 0/0:8,0:8   0/1:3,1:4   1/1:0,4:4
2       136047387    .    T    C    186.27  .       AC=1;AF=0.167;AN=6  GT:AD:DP 0/0:6,0:6   0/0:16,0:16 0/1:4,6:10
2       136048649    .    C    G    127.26  .       AC=1;AF=0.167;AN=6  GT:AD:DP 0/0:13,0:13  0/0:9,0:9    0/1:1,4:5
2       136052318    .    C    T    107.26  .       AC=1;AF=0.167;AN=6  GT:AD:DP 0/0:7,0:7    0/0:13,0:13 0/1:3,3:6
```

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GATK’s best practices for germline short variant discovery

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https://gatk.broadinstitute.org

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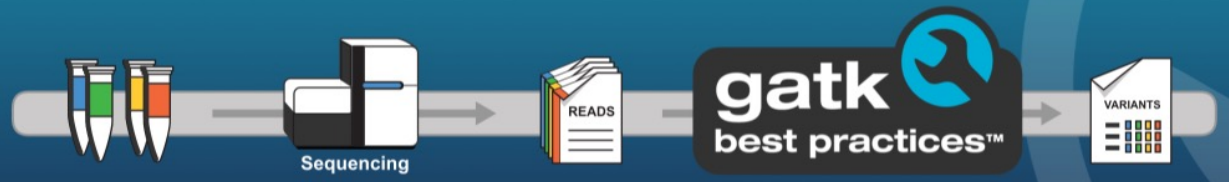
Genome Analysis Toolkit

Variant Discovery in High-Throughput Sequencing Data



Developed in the Data Sciences Platform at the Broad Institute, the toolkit offers a wide variety of tools with a primary focus on variant discovery and genotyping. Its powerful processing engine and high-performance computing features make it capable of taking on projects of any size. Learn more

Find answers to your questions. Stay up to date on the latest topics. Ask questions and help others.

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Best practices, tutorials, and other info to get you started

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Purpose, usage and options for each tool

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Learn more about DRAGEN-GATK

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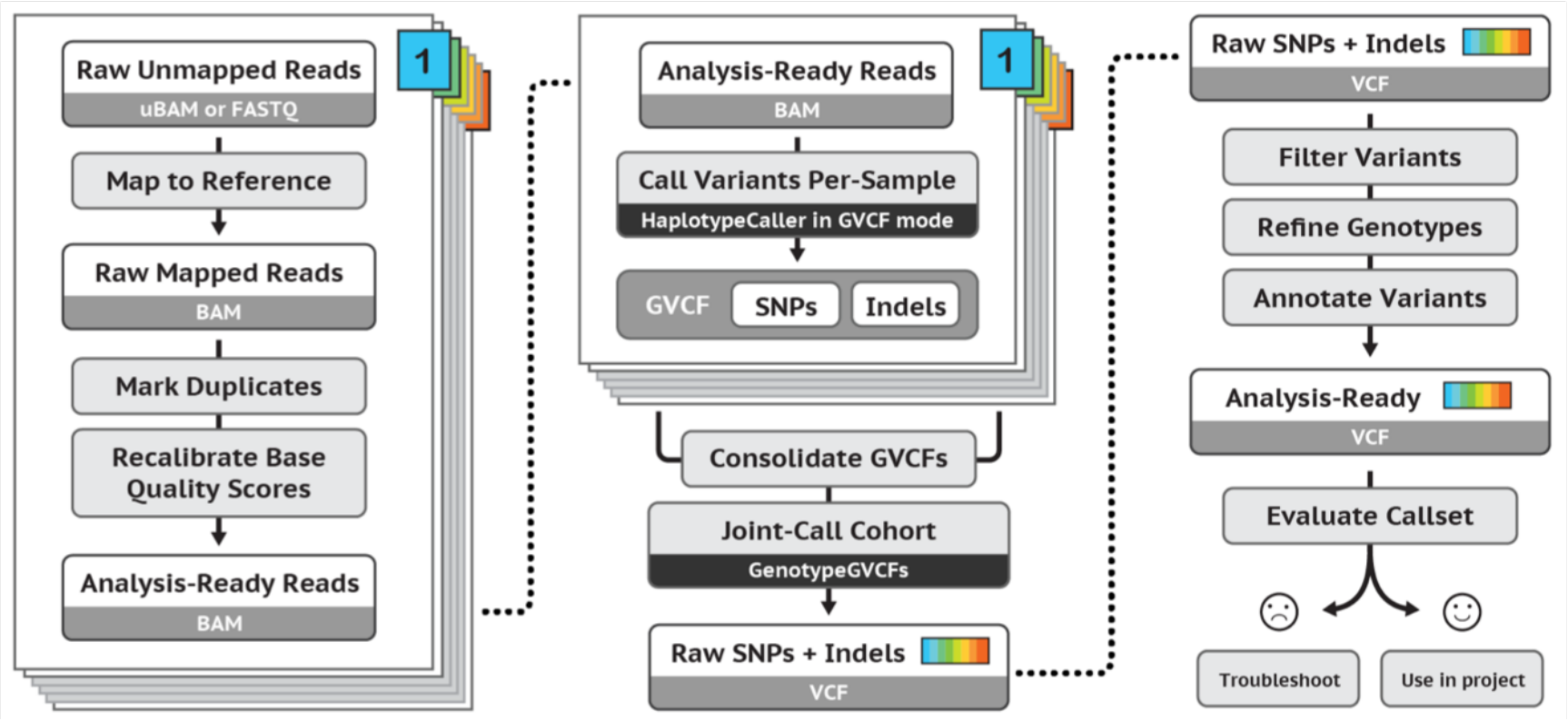
The GATK package download includes all released GATK tools

Run on Cloud

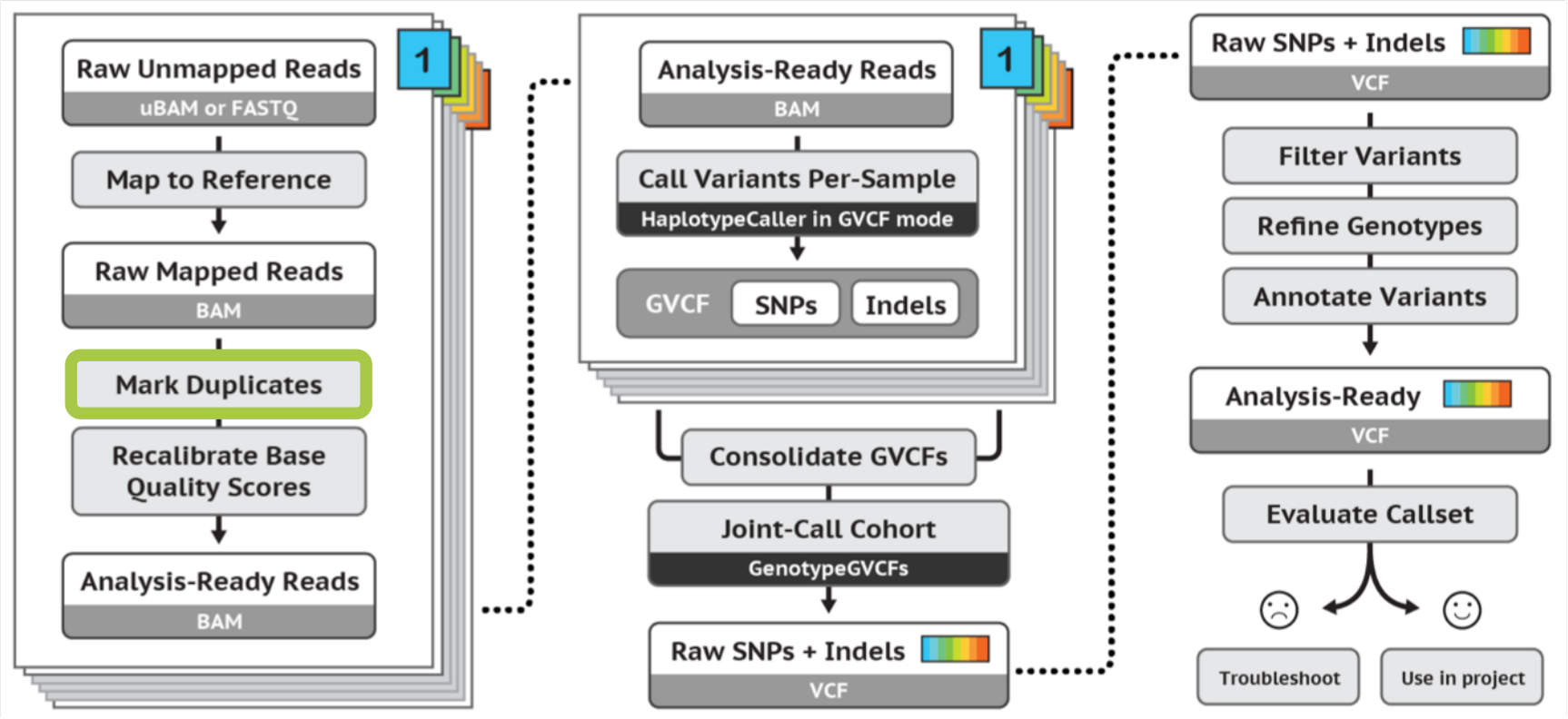
Run on HPC

GATK's best practices workflow for germline short variant discovery



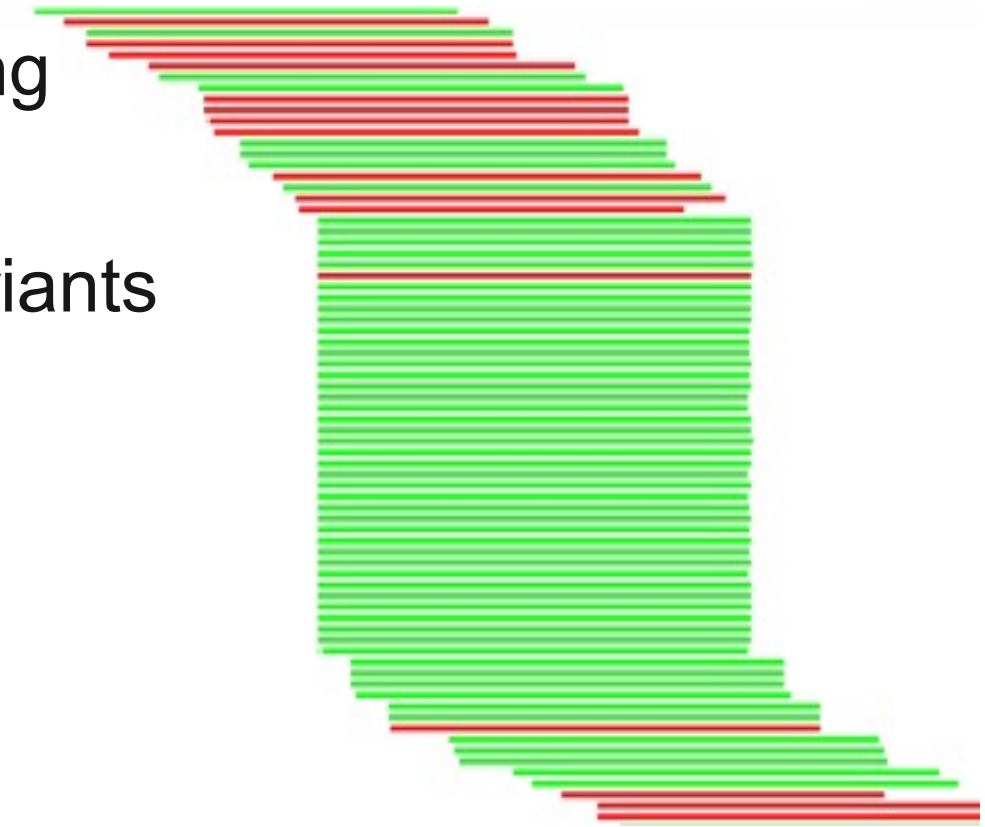
https://software.broadinstitute.org/gatk/best-practices/

Mark Duplicates



Duplicate reads

- PCR duplicates - library preparation
- Optical duplicates - sequencing
- Don’t add unique information
- Gives false allelic ratios of variants
- Should be removed/marked



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MarkDuplicates

GATK / Tool Index / 4.0.1.1

MarkDuplicates (Picard)

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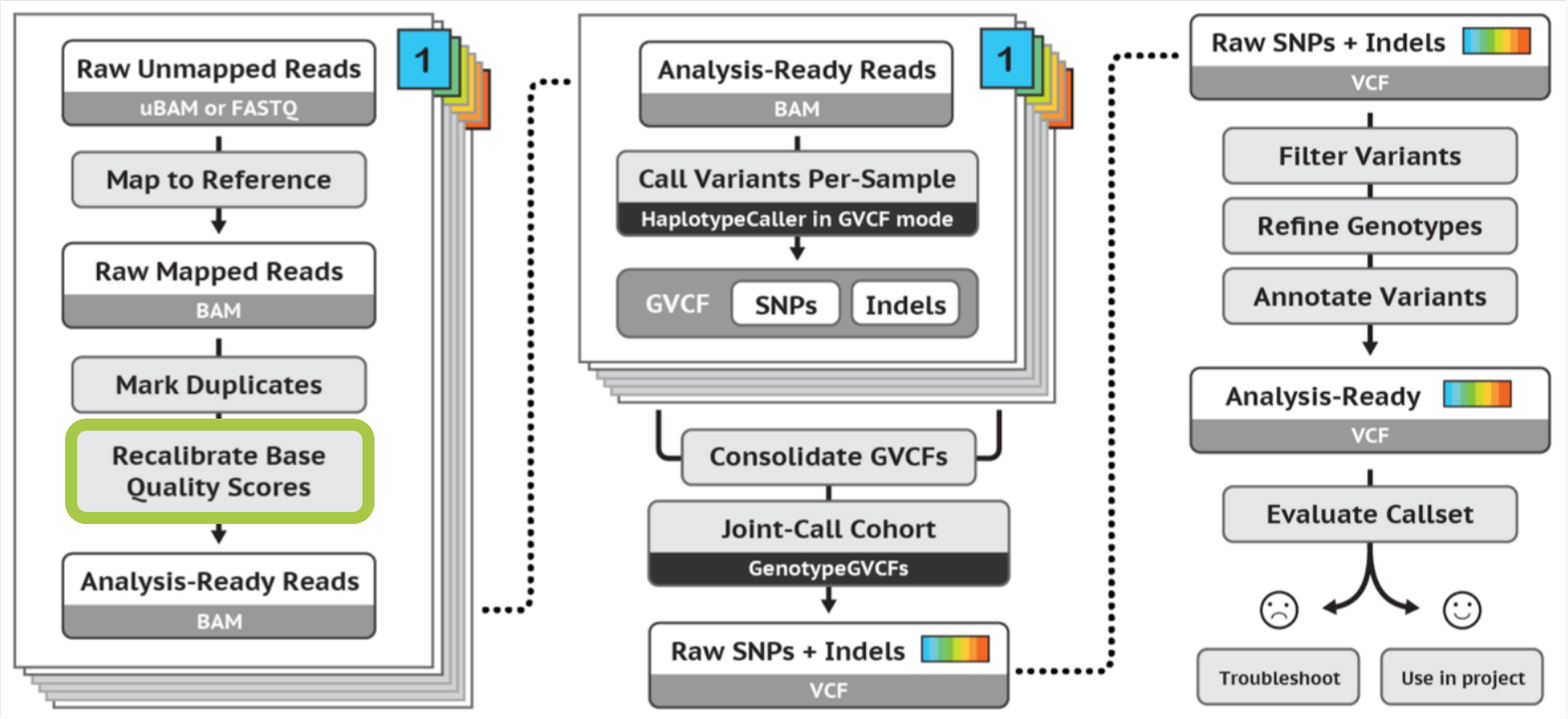
GATK Team
10 months ago · Updated

Identifies duplicate reads.

This tool locates and tags duplicate reads in a BAM or SAM file, where duplicate reads are defined as originating from a single fragment of DNA. Duplicates can arise during sample preparation e.g. library construction using PCR. See also EstimateLibraryComplexity for additional notes on PCR duplication artifacts. Duplicate reads can also result from a single amplification cluster, incorrectly detected as multiple clusters by the optical sensor of the sequencing instrument. These duplication artifacts are referred to as optical duplicates.

```
gatk --java-options -Xmx7g MarkDuplicates \
      -I input.bam \
      -O marked_duplicates.bam \
      -M marked_dup_metrics.txt
```

Base Quality Score Recalibration (BQSR)



Base Quality Score Recalibration (BQSR)

- During base calling, the sequencer estimates a quality score for each base. This is the quality scores present in the fastq files.

- Systematic (non-random) errors in the base quality score estimation can occur.
  - due to the physics or chemistry of the sequencing reaction
  - manufacturing flaws in the equipment
  - etc

- Can cause bias in variant calling

- **Base Qualtiy Score Recalibration** helps to calibrate the scores so that they correspond to the real per-base sequencing error rate (phred scores)

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Base Quality Score Recalibration (BQSR)

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Base Quality Score Recalibration (BQSR) Follow

GATK Team
5 days ago · Updated

BQSR stands for Base Quality Score Recalibration. In a nutshell, it is a data pre-processing step that detects systematic errors made by the sequencing machine when it estimates the accuracy of each base call.

*Note that this **base** recalibration process (BQSR) should not be confused with **variant** recalibration (VQSR), which is a sophisticated filtering technique applied on the variant callset produced in a later step. The developers who named these methods wish to apologize sincerely to anyone, especially Spanish-speaking users, who get tripped up by the similarity of these names.*

Contents

1. Overview
2. Base recalibration procedure details
3. Important factors for successful recalibration
4. Examples of pre- and post-recalibration metrics
5. Recalibration report

Articles in

ActiveRe
(Haploty

Evaluatir
and varia
Mutect2)

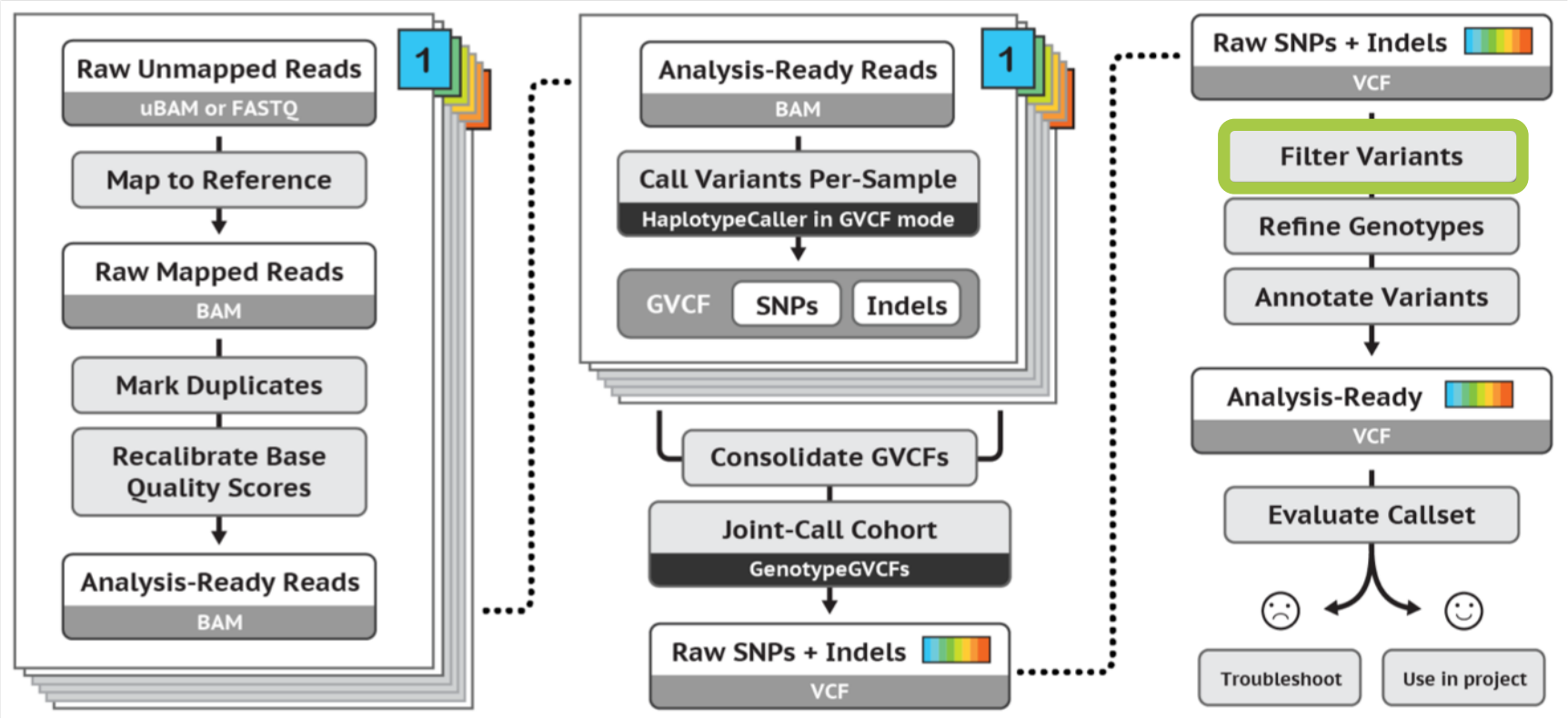
Local re-
determir
Mutect2)

Allele-sp
germline

Variant C

Evaluatir
variant c

Filter variants



https://software.broadinstitute.org/gatk/best-practices/

Germline short variant discovery (SNPs + Indels)

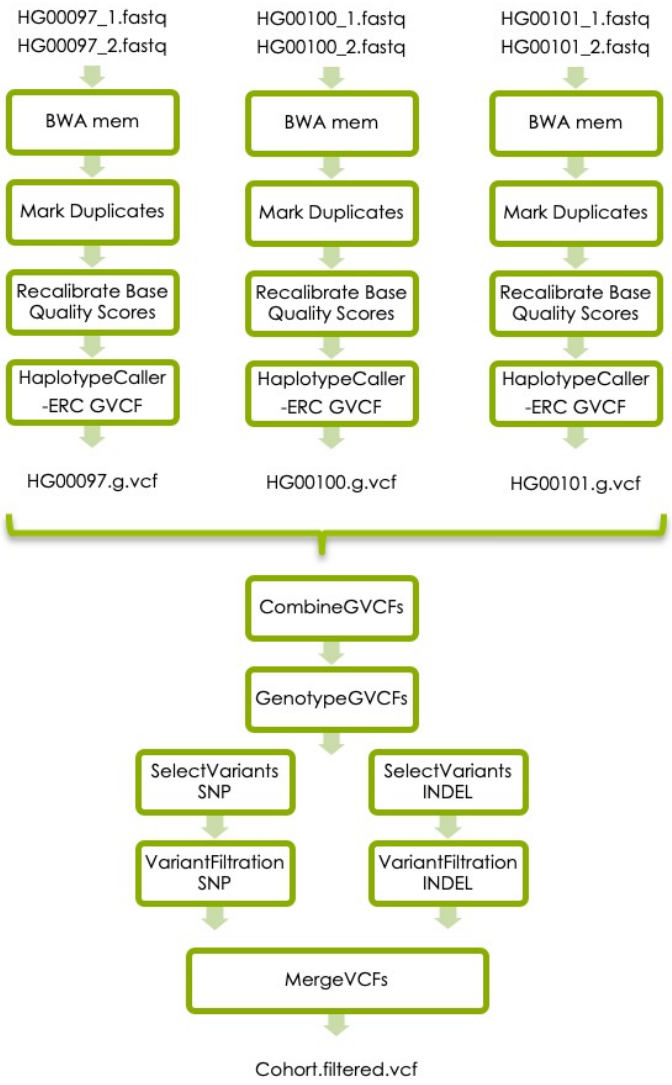
Filtering

- Remove low quality variants

- Variant quality score recalibration (VQSR):
  - For large data sets (>1 WGS or >30WES samples)
  - GATK has a machine learning algorithm that can be trained to recognise ”likely false” variants
  - **We do recommend to use VQSR when possible!**

- Hard filters:
  - For smaller data sets
  - Hard filters on information in the VCF file
  - For example: Flag variants with ”QD < 2” and ”MQ< 40.0”
  - GATK discussion on hard filters: https://gatkforums.broadinstitute.org/gatk/discussion/2806/howto-apply-hard-filters-to-a-call-set

GATK's best practices workflow



More details and links to GATK for each step is found in the lab instructions.

Today’s lab

1000 Genomes data

- Low coverage WGS data
- 3 samples
- Small region on chromosome 2

About the samples:
https://www.internationalgenome.org/data-portal/sample

The Lactase enzyme

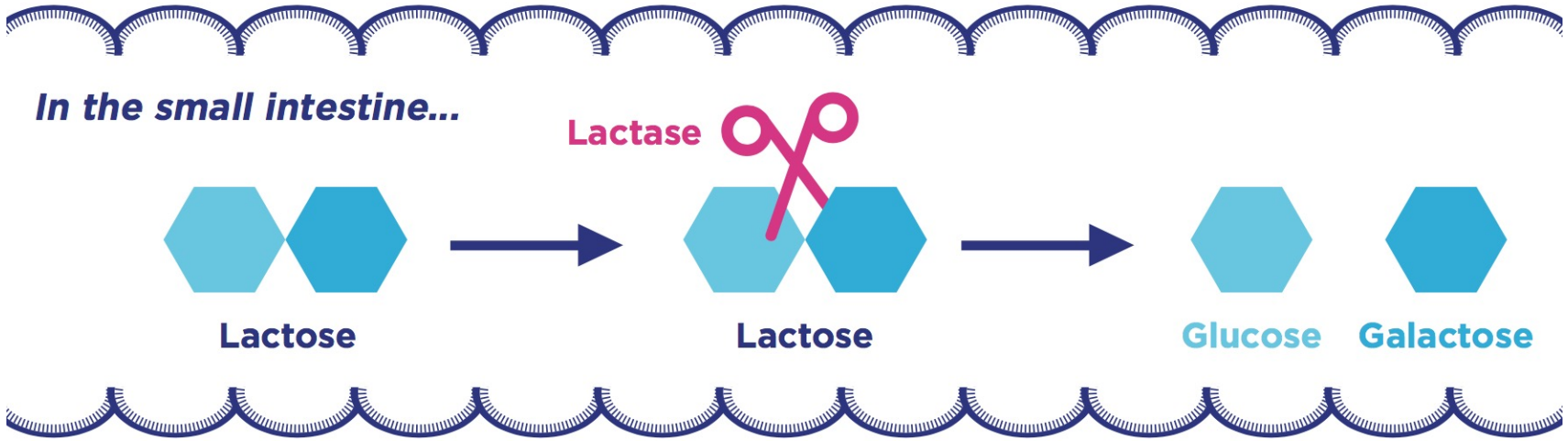


Figure 2. Lactose digestion in the intestine.

- All mammals produce lactase as infants
- Some human produce lactase in adulthood
- Genetic variation upstream of the *LCT* gene cause the lactase persistent phenotype (lactose tolerance)

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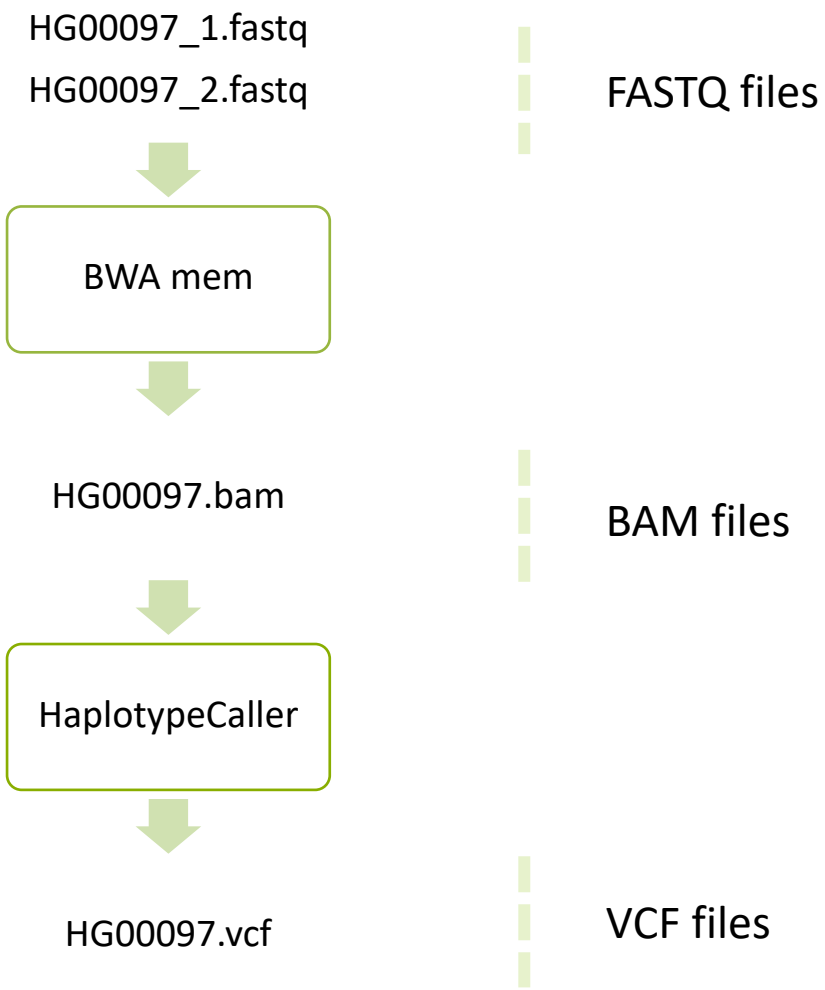
part one:

variant calling in one sample

Basic variant calling in one sample



Part two (if you have time):

variant calling in cohort

Joint variant calling workflow

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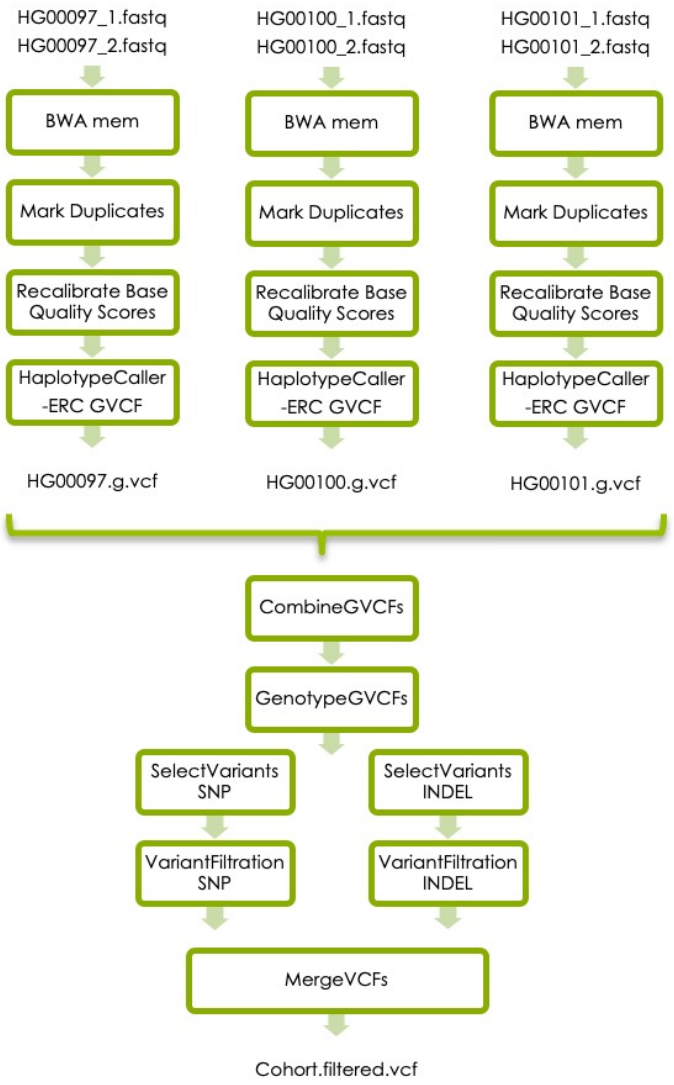
Part three (if you have time):

Follow GATK best practices for short variant discovery

GATK's best practises



First look at video about this linked from schedule!

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https://gatk.broadinstitute.org

gatk

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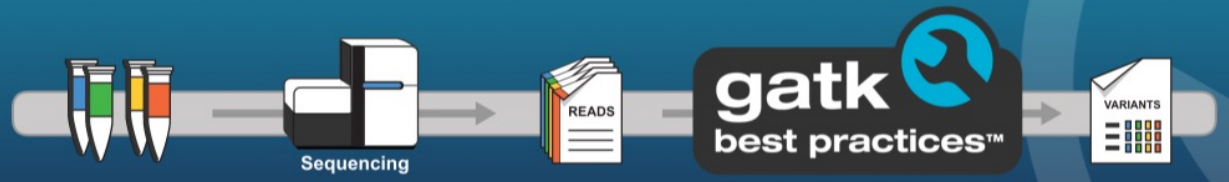
Genome Analysis Toolkit

Variant Discovery in High-Throughput Sequencing Data



Developed in the Data Sciences Platform at the Broad Institute, the toolkit offers a wide variety of tools with a primary focus on variant discovery and genotyping. Its powerful processing engine and high-performance computing features make it capable of taking on projects of any size. Learn more

Find answers to your questions. Stay up to date on the latest topics. Ask questions and help others.

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Questions?